Edge Detection in Noisy Images Using 2D Filtering

ECE 0402 Group Project

April 10, 2026

1 Project Goal

This project studies a narrow signals-and-systems question: why does naive edge detection fail in noisy images, and how can 2D filtering improve it? The project treats an image as a 2D signal and uses convolution, LTI filtering, derivative operators, and smoothing-before-detection as the main conceptual thread.

2 Why This Topic Fits the Course

- The main system operation is 2D convolution.
- Sobel filtering is a direct example of a linear shift-invariant operator on a 2D signal.
- Edge detection highlights high-frequency structure, but noise also lives in that regime.
- The project includes a short demo that can be explained within two minutes.

3 Method

3.1 Image Set

The experiments use three standard images from `skimage.data` and one synthetic image generated in code. The synthetic image is useful because its geometry is fully controlled.

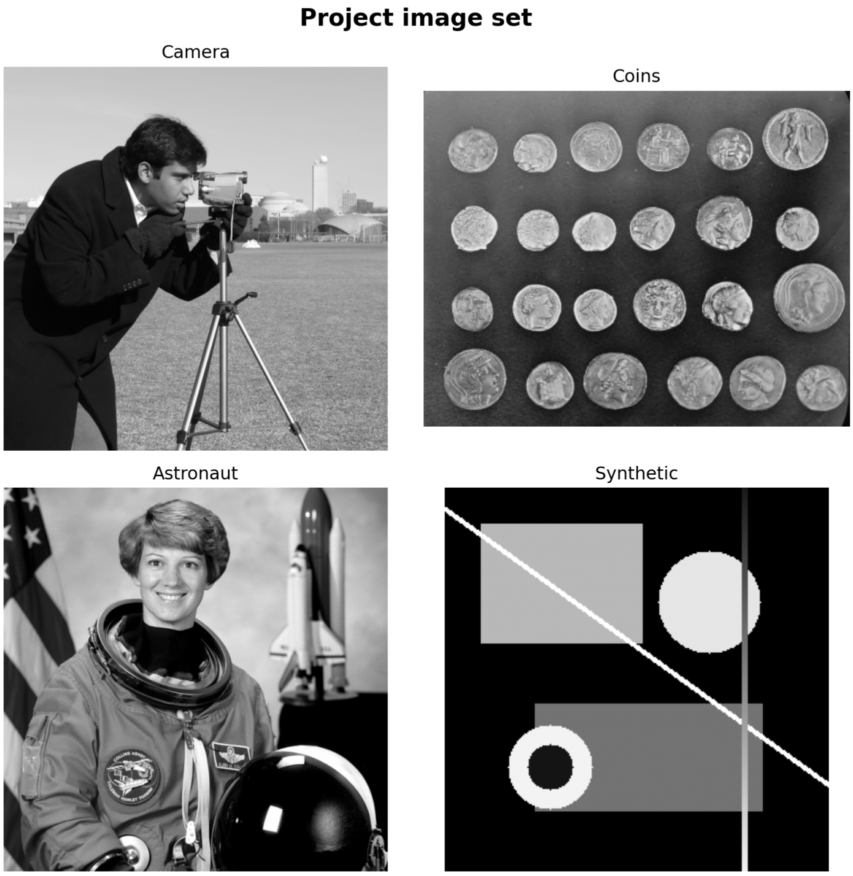

Figure 1: Project image set used in the experiments.

3.2 Pipeline

The comparison pipeline is intentionally simple:

1. apply Sobel directly to the clean image,
2. add Gaussian noise and apply Sobel again,
3. smooth the noisy image with a Gaussian filter before Sobel,
4. compare the result with Canny as a stronger reference pipeline.

The Sobel operator estimates horizontal and vertical intensity changes through two small convolution kernels.

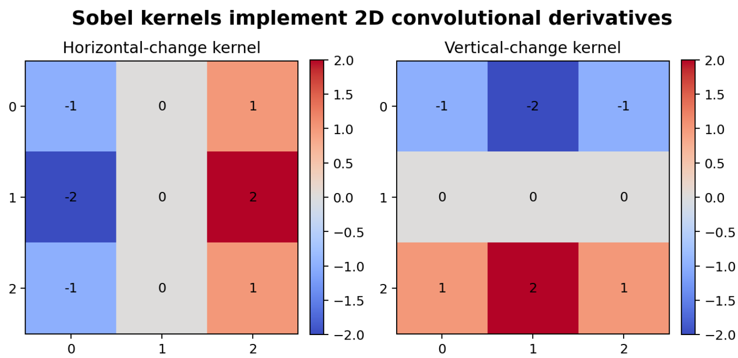

Figure 2: Sobel kernels used to compute local derivatives in two dimensions.

4 Key Result

The core failure mode is easy to see: derivative filters amplify rapid changes, and noise is also a rapid change. This means naive edge detection reacts to random fluctuations as if they were meaningful boundaries.

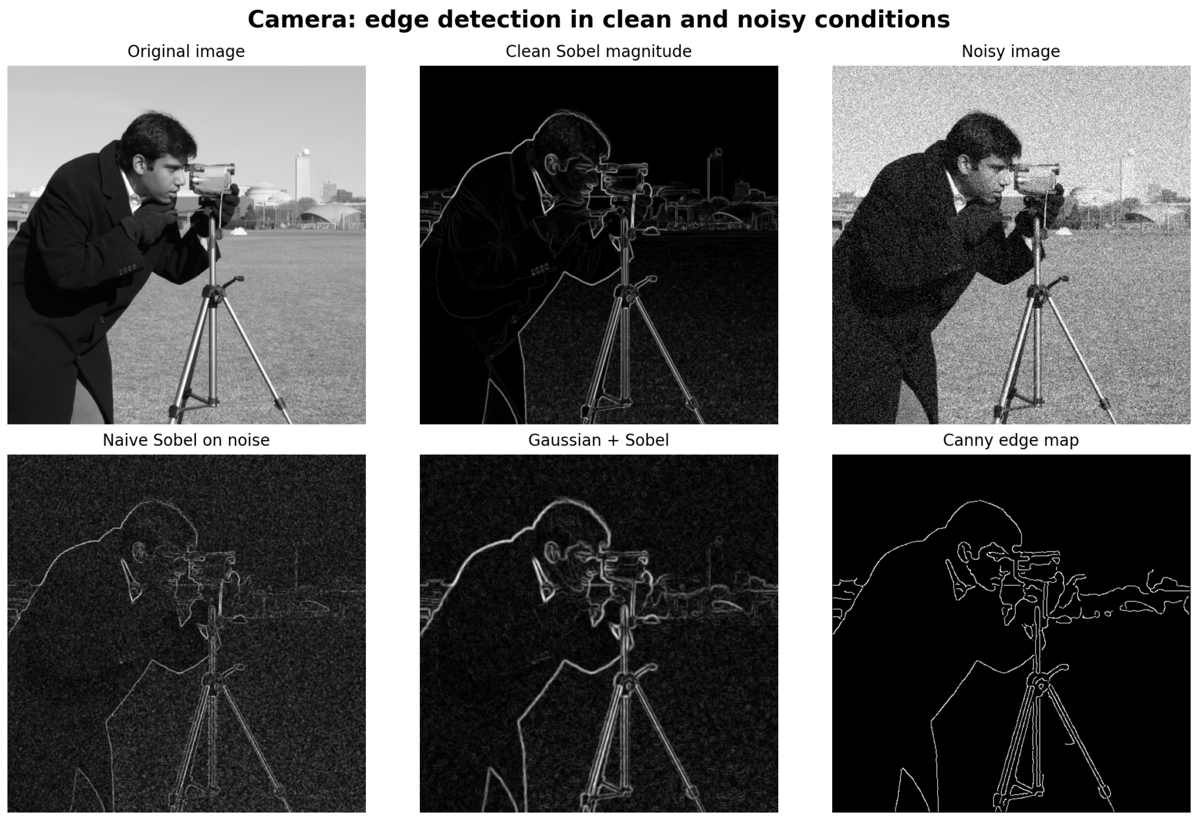

Figure 3: Main demo figure: noise corrupts the naive Sobel response, while smoothing and Canny improve the result.

5 Tradeoff Analysis

Smoothing helps, but too much smoothing weakens fine detail. To make that tradeoff explicit, the project computes an F1 score against a clean-edge reference over a grid of noise levels and smoothing strengths.

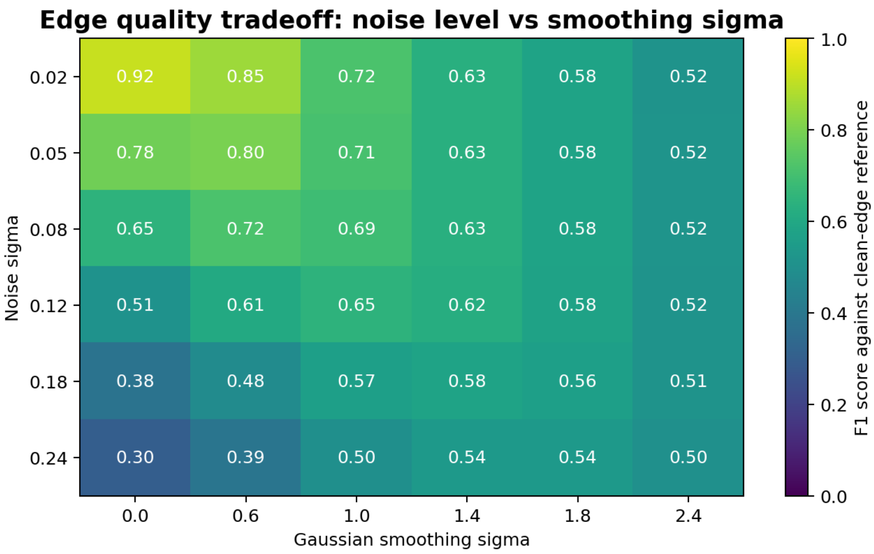

Figure 4: Tradeoff heatmap between noise level and smoothing strength.

The trend is straightforward:

- when noise is small, little or no smoothing preserves the most detail;
- when noise increases, moderate smoothing improves edge quality;
- excessive smoothing eventually removes useful boundaries.

6 Modern Vision Connection

This project does not make CNNs or diffusion models the main story. That would be a category error. The point is to show that modern image models still rest on filtering intuition:

- early CNN filters often learn edge-like or texture-like kernels;
- diffusion models repeatedly denoise an image estimate;
- classical filtering remains the correct conceptual baseline.

7 Two-Minute Demo Script

1. Show the clean image and its Sobel response.
2. Add Gaussian noise and show that naive Sobel now reacts to noise as if it were detail.
3. Smooth the noisy image before Sobel and show the recovery of the main boundaries.
4. End with Canny and explain why a better pipeline still depends on filtering.

8 Team Responsibilities Template

Fill in the names before submission. Do not leave placeholder names in the final version.

Member	Responsibility	Concrete Output
Member A	Project framing and literature check	Topic definition, references, slide story
Member B	Implementation	Python modules, build script, generated figures
Member C	Analysis and validation	Heatmap interpretation, self-check results, rehearsal timing
Member D	Presentation assembly	Slide edits, speaking notes, final rehearsal

Table 1: Template for explicit division of labor.

9 References

1. ECE 0402 course project guideline, `../Project Guideline_ECE0402_2026S.pdf`.

2. scikit-image documentation, "Canny edge detector," https://scikit-image.org/docs/stable/auto_examples/edges/plot_canny.html.

3. scikit-image API documentation, `skimage.filters`, https://scikit-image.org/docs/0.20.x/api/skimage.filters.html.

4. HIPR2, "Sobel Edge Detector," https://homepages.inf.ed.ac.uk/rbf/HIPR2/sobel.htm.

5. HIPR2, "Canny Edge Detector," https://homepages.inf.ed.ac.uk/rbf/HIPR2/canny.htm.

6. B. P. Lathi, *Linear Systems and Signals*, 2nd edition.

7. Luis F. Chaparro, *Signals and Systems Using MATLAB*, 4th edition.